Followup 26-08

Unhelmeted motorcyclist with a subdural hematoma

Patient	55-year-old male. COPD with more than fifty pack-years, insulin-dependent type 2 diabetes, hypertension on atenolol and lisinopril-hydrochlorothiazide, morbid obesity, alcohol use disorder with a history of withdrawal seizures, depression, and multiple prior abdominal surgeries.
Outcome	Survived. **Acute right cerebral convexity subdural hematoma, 12 mm thick, with mild mass effect and 2 mm of leftward midline shift.** Minimally displaced calvarial fracture running from the right parietal bone into the squamous temporal bone, a 3 cm right scalp hematoma, and a nondisplaced right zygomatic arch fracture. No injury anywhere below the neck. The head CT generated a red critical result.
Disposition	No neurosurgical operation. Levetiracetam load, systolic goal under 160, hourly neurologic checks, repeat CT at seven hours showing no significant change. Discharged on hospital day 5, alert and oriented times four, after physical therapy had recommended a comprehensive rehabilitation program.

The call

Dispatched to a motorcycle crash. Fire and police were on scene. He was supine on the ground with a pool of blood beside him and his motorcycle next to him, not wearing a helmet. Police reported he had been travelling roughly 55 mph and had been struck by a car, and that other drivers described him as going too fast. A bystander who identified herself as an anesthesiologist was holding pressure on his head and maintaining cervical spine control until crews arrived.

He was alert but not oriented to the event. He would not tolerate a cervical collar. A BLS crew placed a backboard, and the log roll found nothing on his back. He complained of elbow pain and had cuts and abrasions to both arms. An off-duty advanced EMT from the agency stopped and helped place an 18 gauge IV in each arm.

Course, relative to patient contact:

- **On contact:** Uncontrolled bleeding documented from the head. Two IVs. **Tranexamic acid 1 gram in 100 mL of normal saline.**
- His head bled through the gauze, more was added, and the head was wrapped. **He immediately pulled the bandages off and tried to sit up.** He was redirected repeatedly during the call and reminded more than once that he had been in a motorcycle crash.
- **+8 min:** 158/100, MAP 119, pulse 90, respirations 20, SpO2 97%.
- **+12 min:** Cardiac monitor applied.
- **+16 min:** Left scene. Priority 1.
- **+23 min:** At the receiving facility.

He reported having had a few drinks.

Two things to hold onto. **He kept forgetting why he was there**, and **that one set of vitals.**

Primary impression

Motorcycle crash with head injury and uncontrolled scalp bleeding.

Right on every count, and the crew treated it that way: two large-bore IVs, TXA within minutes, direct pressure, and a priority 1 transport with no delay on scene. Sixteen minutes from contact to moving.

One line in the narrative is worth returning to, because it is the only thing in this chart that reads differently once you know the CT result. **The narrative records that vital signs were checked and were in normal range.** The recorded set was **158/100 with a MAP of 119 and a pulse of 90**, in a 55-year-old who had just been thrown off a motorcycle. That is not a normal blood pressure, and in this particular patient it was not a meaningless one either. It gets its own section below.

Hospital course

Emergency department, day 0. Level 1 trauma activated on mechanism. He arrived confused with a **GCS of 14**, scoring 4 for eye opening, **4 for verbal**, and 6 for motor, which is the numerical version of what the crew wrote down as alert but not oriented. Triage vitals 163/100 with a pulse of 76. He was boarded but had no collar on arrival, and one was placed. He was awake, talking, agitated, restless, protecting his own airway, and repeatedly asking for something to drink. Large right parietal hematoma, road rash to both forearms, and tenderness in the left lower quadrant.

FAST was negative in the cardiac, left upper quadrant, and suprapubic windows and indeterminate in the right upper quadrant, limited by body habitus and cooperation. Chest and pelvis films were unremarkable.

CT head, about ninety minutes after crew contact: a 12 mm acute right convexity subdural hematoma with mild mass effect and 2 mm of leftward midline shift, a minimally displaced fracture from the right parietal bone into the anterior squamous temporal bone, a 3 cm scalp hematoma, and a nondisplaced right zygomatic arch fracture. The radiologist called it in as a red critical result. CT of the cervical spine, chest, abdomen, and pelvis found no acute traumatic injury anywhere.

Neurosurgery recommended no operation, a levetiracetam load for seizure prophylaxis, a systolic blood pressure goal below 160, and a repeat CT in the morning. He was given levetiracetam, tetanus vaccine, and fentanyl.

His scalp kept bleeding. A compressive wrap went on, and about an hour later the wound was actively bleeding and hard to assess through an enlarging hematoma. It was infiltrated with lidocaine and epinephrine and closed with two sutures, a figure-of-eight and a simple interrupted, which controlled it. **He removed the compressive dressing multiple times.** He was placed in wrist restraints, and the collar was cleared clinically about three hours after arrival on a negative examination.

Hospital day 1. Repeat CT at seven hours: essentially unchanged, subdural now measured at 11 mm. Hourly neurologic checks. Chemical thromboprophylaxis was held because of the intracranial hemorrhage, with compression devices instead. He was placed on a CIWA protocol for alcohol withdrawal with thiamine, folate, and multivitamin. He desaturated overnight and went from 2 to 4 L. Neurosurgery and critical care both signed off with the bleed stable, planning a repeat CT in two to four weeks.

Days 1 through 5. Wound care shaved and cleaned the scalp, revealing an open wound and hematoma measuring roughly 1.4 by 3 by 1.5 cm, packed with hypochlorous acid gauze, and dressed the road rash on both arms with a silver contact layer. **Physical therapy scored him at Rancho Los Amigos level VI, confused and appropriate, requiring moderate assistance**, walking 16 feet with a front-wheeled walker against a baseline of full independence, and recommended a comprehensive rehabilitation program. He was discharged on hospital day 5, alert and oriented times four.

Final diagnoses: acute right cerebral convexity subdural hematoma with mass effect and midline shift; calvarial fracture; right zygomatic arch fracture; large scalp hematoma with laceration; closed head injury; bilateral upper extremity abrasions; alcohol use disorder; acute metabolic encephalopathy, improving.

Education points

What a subdural hematoma actually is

Three membranes wrap the brain. The **dura mater** is the tough outer layer stuck to the inside of the skull. The **arachnoid** is a delicate middle layer. The **pia** clings directly to the brain surface. A subdural hematoma is blood collecting between the dura and the arachnoid.

The bleeding comes from **bridging veins**. Veins draining the surface of the brain have to cross the gap from the moving brain to the fixed dural sinuses in order to empty. That crossing is where they are vulnerable, because one end is anchored to something that moves and the other to something that does not. Take a head travelling at 55 mph and stop it against pavement, and the brain keeps going for a fraction of a second longer than the skull does. The bridging veins are what get stretched and torn in that fraction of a second.

Because the source is venous, the bleeding is slower than an epidural, which is usually arterial and can kill in an hour. On CT a subdural is **crescent shaped and follows the contour of the brain, and it crosses suture lines** because the dura is not fixed at the sutures the way the outer membrane in an epidural is. That is how a radiologist tells them apart.

Now the part that matters clinically: the skull is a closed box. It contains brain, blood, and cerebrospinal fluid, and the total volume is fixed. Add a 12 mm collection of blood and something has to give. Initially cerebrospinal fluid is displaced and venous blood is squeezed out, which is why a patient can look nearly normal at first. Once that compensation is exhausted, pressure inside the skull rises quickly, and the brain is pushed away from the mass. **The 2 mm of leftward midline shift on this CT is that push, measured.** It is a small number and it is not nothing, because it means the compensating room had already started to run out.

He was not operated on, and understanding why is as useful as understanding the injury. He had a GCS of 14 rather than a rapidly falling one, mild rather than significant compression, and a repeat scan at seven hours showing no growth. Surgery for an acute subdural is a decision about trajectory as much as about size, which is why the plan was hourly neurologic checks and a second scan rather than an operating room.

> **Further reading on subdural hematoma.** The surgical thresholds, the chronic and subacute forms, and the complications go further than a case document can carry. StatPearls has a free chapter: https://www.ncbi.nlm.nih.gov/books/NBK532970/

The blood pressure, and why nobody brought it down

Here is the vital sign again: **158/100, MAP 119, pulse 90.** At triage, 163/100 with a pulse of 76.

Start with the equation that governs everything about a head injury. **Cerebral perfusion pressure equals mean arterial pressure minus intracranial pressure.** The brain is perfused by the difference between the pressure pushing blood in and the pressure inside the skull resisting it. When a subdural hematoma raises intracranial pressure, the only variable the body can move to keep blood flowing is the mean arterial pressure, so it raises it.

Hypertension in a head-injured patient is frequently the body defending its own brain, and dropping it can convert a compensated injury into an ischemic one.

That is why the neurosurgery instruction was a systolic **below 160** rather than a normal pressure. High enough to perfuse, low enough not to extend the bleed. It is a ceiling, not a target.

The pattern to know by name is the **Cushing reflex**: rising blood pressure, falling heart rate, and an irregular breathing pattern, driven by the brainstem making a last effort to push blood past a pressure it is losing to. It is a **late** sign and it means herniation is close. This patient did not have it. His pressure was up and his rate was 90, then 76, and his breathing was regular, and he never developed the full triad.

Two honest qualifications, because this is the kind of reasoning that gets overextended. **He takes atenolol twice a day at home**, a beta blocker, which caps the heart rate response and makes any rate on this chart weaker evidence than it looks. And pain, fear, and alcohol all raise blood pressure without any help from the brain. **The pressure alone does not prove raised intracranial pressure.** What it does is refuse to be normal, and calling it normal in the narrative is what closes off the thinking that follows from it.

Agitation is a neurologic finding

This is the single most portable thing in this case.

He would not tolerate a collar. He pulled his own bandages off within seconds of them being applied. He tried to sit up. He had to be redirected repeatedly and reminded more than once what had happened to him. In the emergency department the same behavior continued, and he removed his compressive dressing several times and ended up in wrist restraints. Physical therapy scored him the next day at **Rancho Los Amigos level VI**, which is the formal name for confused but appropriate and needing moderate assistance.

None of that is a difficult patient. **All of it is the injury.** Diffuse brain injury and rising intracranial pressure produce restlessness, combativeness, and impaired new memory formation, and the memory failure is why reorientation does not stick. Telling a patient with a fresh subdural what happened to him is not a conversation, it is a test, and needing to repeat it is the result.

The differential for the acutely agitated trauma patient is short and every item on it will kill:

- **Hypoxia.** Always first, always free to check.
- **Hypoglycemia.** Free to check, and instantly reversible.
- **Hypoperfusion.** An agitated patient may be an under-perfused brain.
- **Head injury.** This patient.
- **Pain.** Real, and the last one to assume.
- **Intoxication or withdrawal.** The one everybody reaches for, and the one to reach for last.

The practical consequence is that **agitation should be charted like a vital sign, with a time next to it and a trend.** An agitation that is getting worse is a neurologic examination getting worse, and it may be the only sign you get before the pupils change.

Prehospital considerations

He met protocol criteria for a TBI patient, and it happened before anybody saw a CT. Berrien County MCA protocol 2-12, Head Injury, applies to any patient with a mechanism consistent with brain injury plus **any altered mental status, defined as a GCS below 15**. He was not oriented to the event on scene and scored 14 in the emergency department. He met it on arrival. The protocol names four things to prevent, and they are worth memorizing because they are the four that turn a survivable brain injury into a worse one: **hypoxia, hyperventilation, hypotension, and hemorrhage.** The full binder is at http://www.berrienmca.org/protocols/.

Trend the blood pressure in a head injury. Protocol 2-12 sets an adult systolic target above 110 and says plainly not to wait for the patient to become hypotensive before acting. His pressure was never near that floor and was never the problem. The reason to keep taking it is that in a head injury the blood pressure is the number most likely to move and the one you most need a direction on, because cerebral perfusion pressure depends on it and there is no other window into what the brain is getting.

Untolerated cervical collar, and the right call. He refused it, the crew did not force it, and a collar was not placed until he reached the emergency department. Protocol 2-6, Spinal Injury Assessment, makes his assessment positive on three separate criteria: altered mental status, use of intoxicants, and a distracting injury, so spinal precautions were indicated and were maintained.

The stabilization was never the collar. It was the coordinated log roll, the board, and the fact that his head was held by hand from before the crew arrived until he was packaged. A rigid collar limits gross range of motion, and it also compresses the soft tissues of the neck including the jugular veins. Impeding venous drainage from the head raises intracranial pressure, and fighting a collar onto a combative head-injured patient raises it further through the struggle itself. He was manually stabilized the entire time, he had no cervical injury, and forcing the issue would have cost him something and bought nothing. That was good judgment and it should be recognized as such.

Scalp bleeding is its own problem. The scalp has a dense blood supply and its vessels are held open by the fibrous tissue they run through, so they do not retract and clamp down the way vessels elsewhere do. A scalp laceration in a patient with normal clotting can produce genuinely dangerous blood loss and it will keep going under a loose dressing. His hemoglobin fell from 15.0 to 12.8 in a day with a clean chest, abdomen, and pelvis CT and a negative FAST. **The only place he was losing blood was his head**, and the emergency department eventually had to inject lidocaine with epinephrine and throw two sutures to stop it. Direct, firm, focal pressure over the bleeding point beats a thick wrap every time.

Alcohol never explains the neurologic examination. He told the crew he had a few drinks, and his level came back at **33**, well under the legal limit for driving. That number would not make anybody forget what had happened to them twenty minutes ago. The rule is simple and it does not have exceptions worth learning: in a head-injured patient, alcohol is a comorbidity and not a diagnosis, and every abnormal finding belongs to the head until imaging says otherwise.

Mechanism from third parties. Police had the speed, the fact that he was struck by a car, and witness accounts of how he was riding. A bystander had been holding pressure and c-spine before anybody arrived and knew what his level of consciousness had been doing. **That information exists on scene for a few minutes and then disperses.** The receiving trauma team activated a level 1 on mechanism, which means the speed the crew relayed was doing real work.

The interventions

PREHOSPITAL INTERVENTIONS

Two 18 gauge IVs, one in each arm. Correct for a level 1 trauma patient, and worth understanding why the gauge matters more than the number of lines. Flow through a tube rises with the fourth power of its radius and falls with its length, so a short fat catheter moves fluid enormously faster than a long thin one. Two large-bore peripheral lines are the standard opening position for a patient who may need blood, and getting them before the patient becomes hard to stick is the whole point.

Tranexamic acid, 1 gram in 100 mL. TXA blocks the binding site on plasminogen that lets it attach to fibrin, which prevents the conversion to plasmin that would otherwise dissolve a clot. It does not make clots. It keeps the ones already formed from being broken down, and its benefit is time dependent, which is why it belongs to prehospital care rather than the hospital.

The evidence has two halves. **CRASH-2** established a mortality benefit in bleeding trauma patients when given early, with the benefit shrinking and reversing as the delay grew. **CRASH-3** studied it specifically in traumatic brain injury and found a benefit in mild to moderate injury when given within three hours, with no benefit in the most severe cases. This patient sat in the middle of both: an actively bleeding scalp and a head injury with a GCS of 14. He got it within minutes of contact.

Protocol 2-12 directs the crew to consider TXA where hemorrhage is present, routing the criteria through protocol 2-14, Hemorrhagic Shock. It also carries a line worth knowing: **contact medical control for patients who do not meet the clinical criteria for TXA but in whom hemorrhage is suspected.** This patient was bleeding and was not in shock, with a systolic of 158, which is exactly the situation that line was written for.

Spinal motion restriction and the log roll. Discussed above. The log roll also served a second purpose that is easy to forget, which is that it is the only look anybody gets at the back of a supine patient, and the crew documented that they took it and found nothing.

Direct pressure and wrapping. Attempted, defeated by a scalp that kept bleeding through, and then defeated again by a patient who pulled the dressing off. This is not a failure of technique. It is what an uncontrolled scalp bleed on an agitated patient looks like, and the same thing happened twice more inside the hospital.

Sixteen minutes on scene, priority 1. With a level 1 mechanism, an obvious head injury, and a bleed nobody could see, the correct intervention was the truck moving. It did.

IN-HOSPITAL INTERVENTIONS

The CT panel. Head, cervical spine, and chest, abdomen and pelvis with contrast, plus reconstructions of the thoracic and lumbar spine. In a level 1 trauma with an unclear examination this is not shotgunning, it is the only way to exclude injuries a confused patient cannot report. Everything below the neck was clean, which is the finding that made the falling hemoglobin interpretable.

FAST examination. Bedside ultrasound looking for free fluid in four windows: around the heart, in the right upper quadrant between liver and kidney, in the left upper quadrant between spleen and kidney, and behind the bladder. It answers one question quickly, which is whether there is blood in a body cavity. Three windows were negative and the right upper quadrant was **indeterminate**, limited by body habitus and cooperation, and the physician documented that rather than calling it negative. **An indeterminate FAST is not a negative FAST**, and the CT is what actually cleared his abdomen.

Levetiracetam load. A seizure after a traumatic brain injury raises cerebral metabolic demand and intracranial pressure in a brain that can afford neither, so anticonvulsant prophylaxis is standard for significant intracranial hemorrhage. Levetiracetam binds a protein on synaptic vesicles and reduces excitatory neurotransmitter release, and it is chosen because it is available intravenously, has almost no drug interactions, needs no levels, and does not drop blood pressure.

Systolic goal below 160 with hourly neurologic checks. The two halves of the same plan. The pressure ceiling protects the clot from being extended, and the hourly examination is the monitor, because there is no continuous measurement of intracranial pressure in a patient who has not had a bolt placed. **In a head injury the neurologic examination is the vital sign**, and repeating it on a schedule is how a deterioration gets caught.

Chemical thromboprophylaxis held. He is immobile, obese, and injured, which is close to a perfect setup for venous thromboembolism, and he also has an active intracranial bleed. Anticoagulating him could extend it. The compromise was mechanical compression devices, which reduce venous stasis without touching the clotting cascade, and the competing risk was reordered in time rather than dismissed.

The CIWA protocol. He drinks, has a documented history of withdrawal seizures, and was about to spend several days in a hospital not drinking. Withdrawal seizures in a patient with a fresh subdural are a genuinely dangerous combination, and scoring him on a schedule with medication available is how that gets pre-empted rather than discovered.

The labs, and what they mean

Reference ranges vary between institutions and analyzers. These are the ranges printed on the source reports.

Coagulation and blood count

Lab	Day 0	Day 0, later	Day 1	Reference range
Prothrombin time (s)	10.3	not repeated	not repeated	11.0 to 13.5
INR	1.0	not repeated	not repeated	0.8 to 1.2
Activated partial thromboplastin time (s)	23.2	not repeated	not repeated	24.0 to 34.0
Hemoglobin (g/dL)	15.0	13.9	12.8	12.0 to 16.0
Hematocrit (%)	46.5	42.2	39.3	37.0 to 47.0
Platelet (x10^9/L)	178	173	185	150 to 400
White blood cell (x10^9/L)	12.4	18.6	16.1	4.0 to 10.8
Neutrophils, absolute (x10^9/L)	8.11	not reported	not reported	1.80 to 7.80

The coagulation studies are the first thing anybody checks in a head bleed, and his were clean. A patient with a subdural hematoma who cannot clot has a fundamentally different injury from one who can, because the same 12 mm collection will keep growing. His prothrombin time and INR were normal and his **aPTT was 23.2, below the bottom of the range**, which is a mildly shortened clotting time rather than a prolonged one. That happens in acute injury and acute illness as clotting factors surge, and it is the opposite of the problem the test is looking for.

The hemoglobin is the more instructive number. It went 15.0, then 13.9, then 12.8 across a day, which is a real drop. His FAST found nothing, his chest, abdomen, and pelvis CT found no acute injury, and his pelvis film was clean. There was no cavity to bleed into. **The only place he lost that blood was his scalp**, plus dilution from fluids.

The teaching underneath that is worth stating flatly. **Hemoglobin is a concentration, not a volume.** When someone bleeds, they lose red cells and plasma in the same proportion, so the concentration does not move until interstitial fluid shifts in or somebody gives crystalloid, and that takes hours. His first hemoglobin was 15.0 and entirely normal, taken from a man who had visibly pooled blood on the pavement. **An early normal hemoglobin in an actively bleeding patient tells you almost nothing.**

The white count went from 12.4 to 18.6 and back toward normal without antibiotics, which is the reactive leukocytosis of major injury rather than infection.

Chemistry, gas, and toxicology

Lab	Day 0	Day 1	Reference range
Bicarbonate (mmol/L)	18	22	22 to 29
Anion gap (mmol/L)	14	11	5 to 14
Lactic acid (mmol/L)	3.2	not repeated	0.0 to 2.0
PCO2, venous (mmHg)	30	not repeated	41 to 51
Glucose (mg/dL)	282	166	70 to 99
Potassium, serum, hemolyzed specimen (mmol/L)	4.4	3.7	3.5 to 5.1
Potassium, point of care (mmol/L)	2.7	not repeated	3.5 to 4.5
Ionized calcium (mmol/L)	1.14	not repeated	1.15 to 1.33
AST (U/L)	68	25	0 to 50
Ethanol (mg/dL)	33	not repeated	Less than 20
hs troponin T, 2 hour (ng/L)	46	not repeated	Less than 15

The potassium is the finding worth the most on your next shift, and it is a lesson about specimens rather than about this patient. Two potassium results were drawn within minutes of each other. The serum value came back at **4.4**, comfortably normal, on a specimen the laboratory flagged as **hemolyzed**. The point of care value from the blood gas came back at **2.7**.

Hemolysis means red cells ruptured in the tube, usually from a difficult draw, a small catheter, excessive suction, or rough handling. Red cells are packed with potassium at roughly thirty times the concentration of the plasma around them. **When they burst, that potassium spills into the sample, and the reported serum potassium goes up.** A hemolyzed specimen therefore cannot rule out hypokalemia; it can only hide it. His subsequent clean draws came back at 3.7 and 3.7, and the team treated the 2.7.

The same hemolysis note attaches to his bicarbonate, his AST, and his ethanol on that panel. **When a laboratory tells you a specimen was hemolyzed, it is telling you which direction the errors run**, and potassium is the one where getting it backward matters most.

Ethanol 33. Below the legal limit for driving in every state. Whatever else was wrong with his mental status, that number was not the cause of it.

Lactate 3.2 with a bicarbonate of 18 and a venous PCO2 of 30. A metabolic acidosis with the body compensating by breathing it off, and he was documented at 19 to 20 breaths a minute doing exactly that. The lactate is the interesting part, because **he was never hypotensive at any point in this record.** The intensive care team attributed it to his alcohol use and possible thiamine deficiency rather than to shock. Lactate rises from tissue that is not getting enough oxygen, and it also rises from catecholamines, from the work of breathing, from alcohol, and from thiamine deficiency blocking the enzyme that moves pyruvate into the mitochondria. **A raised lactate means look harder. It does not by itself mean shock.**

Glucose 282 falling to 166, in a man with insulin-dependent diabetes under a maximal stress response. Real diabetes plus stress hyperglycemia, and neither one needed anything acutely.

Two values worth reading on your own. His **troponin was 46 with a delta of 27** at two hours, which is a rising troponin that nobody pursued and nobody needed to, because a catecholamine surge and blunt chest trauma both produce exactly that without any coronary event. His **ionized calcium was 1.14**, just below the floor, before he received any blood products. Calcium is required for the clotting cascade, where it is designated factor IV, and low ionized calcium on arrival in a bleeding trauma patient is associated with greater transfusion requirement and worse outcomes. Neither needed action here.

Learning points

1. **Agitation in a trauma patient is a neurologic finding until proven otherwise.** He refused a collar, pulled his dressings off, tried to sit up, and had to be told repeatedly what had happened to him, and the CT found a 12 mm subdural with midline shift. Hypoxia, hypoglycemia, hypoperfusion, and head injury all come before intoxication on that list. Chart agitation with a time next to it, because a worsening agitation is a worsening neurologic examination.
2. **Do not chase a head-injured patient's blood pressure down.** Cerebral perfusion pressure is mean arterial pressure minus intracranial pressure, so when the pressure inside the skull rises the body raises the pressure outside it to keep blood moving. Neurosurgery set a ceiling of 160 systolic rather than a normal target for exactly that reason. A 158/100 with a MAP of 119 in a fresh head injury is a finding, not a normal set.
3. **Alcohol is a comorbidity, never an explanation.** His level was 33, under the legal limit, in a man who could not retain what had happened to him. Every abnormal neurologic finding in a head-injured patient belongs to the head until imaging says otherwise.
4. **The first hemoglobin in a bleeding patient is close to useless.** His was 15.0 with blood pooled on the pavement beside him, and it fell to 12.8 over the next day with a completely clean chest, abdomen, and pelvis. Concentration does not fall until dilution catches up with the loss, which takes hours. Treat what you can see, not what the number says.
5. **A hemolyzed specimen hides a low potassium.** Red cells hold roughly thirty times the potassium of the plasma around them, so rupturing them in the tube pushes the reported value up. His hemolyzed serum read 4.4 and the point of care sample read 2.7 minutes apart. When a result carries a hemolysis flag, know which way the error runs before you trust it.

Prepared from records obtained through the follow-up request process and de-identified for education. To request follow-up on a call you ran, fill out the request form at https://bcmca.github.io/followup/